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基于多智能体强化学习的重载列车群组协同控制

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摘要：为了探索提升重载铁路运输系统效率的新方法，本文研究了由多列独立重载列车组成运营编组的智能协同控制问题。其核心挑战在于：在严格的安全约束下，于宏观运行层面实现多列车的速度同步、间距保持以及整体能效优化。针对该问题的高维、非线性及分布式决策特征，本文提出了一种完全数据驱动的多智能体强化学习框架。该框架摒弃了对精确解析模型的依赖，将每列车建模为一个智能体，通过与环境的不断试错交互来自主学习协同控制策略。本文首先建立了详细的重载列车纵向动力学模型，将其作为智能体交互的高保真仿真环境。此外，设计了以多智能体近端策略优化（MAPPO）算法为核心的训练框架，并辅以基于势场的安全引导奖励和多目标优化奖励。仿真实验表明，与标准的多智能体近端策略优化基准相比，所提出的数据驱动框架在确保零安全违规的前提下，能够显著提高列车组的速度同步精度，并降低整体牵引能耗。

关键词：重载列车；列车组协同控制；数据驱动控制

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Cooperative Control for Heavy-Haul Train Groups Based on Multi-Agent Reinforcement Learning

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Abstract: To explore novel methodologies for enhancing the efficiency of heavy-haul railway transport systems, this paper investigates the intelligent cooperative control problem of multiple independent heavy-haul trains forming an operational group. The core challenge lies in achieving speed synchronization, distance maintenance, and overall energy efficiency optimization of multiple trains at the macroscopic operation level under strict safety constraints. Addressing the high-dimensionality, nonlinearity, and distributed decision-making characteristics of this problem, this paper proposes a fully data-driven multi-agent reinforcement learning framework. Abandoning reliance on precise analytical models, each train is modeled as an agent that autonomously learns cooperative control strategies through continuous trial-and-error interaction with the environment. This paper first establishes a detailed longitudinal dynamics model of heavy-haul trains as a high-fidelity simulation environment for agent interaction. Furthermore, a training framework is designed centered on the Multi-Agent Proximal Policy Optimization algorithm, supplemented by potential-field-based safety guidance rewards and multi-objective optimization rewards. Simulation experiments demonstrate that compared with the standard Multi-Agent Proximal Policy Optimization baseline, the proposed data-driven framework can significantly improve the speed synchronization accuracy of the group and reduce overall traction energy consumption while ensuring zero safety violations.

Key words: heavy-haul train; train group cooperative control; data-driven control

0 Introduction

Heavy-haul railway transportation serves as the core carrier within the bulk cargo logistics system, where its transportation efficiency is directly correlated with the stability of energy and raw material supply chains [1]. To further exploit line capacity and reduce operational costs, the operational mode of heavy-haul trains is progressively evolving from traditional single-train operations toward Virtual Coupling and automatic driving technologies based on Vehicle-to-Vehicle (V2V) communication [2]. However, a heavy-haul train is intrinsically a high-dimensional multi-mass system characterized by significant time delays and strong nonlinearities[3], subject to constraints imposed by locomotive traction forces, complex nonlinear coupler forces, and air braking systems with pronounced hysteresis characteristics [1, 4]. In high-density Virtual Coupling scenarios, a fundamental contradiction exists between the response lag of actuators and the demands for high-frequency cooperative control. If control strategies fail to adequately account for the aforementioned dynamic constraints, severe safety incidents—such as longitudinal impulses, coupler breakage, or even derailment—may easily

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occur [1].

Faced with these complex dynamic challenges, existing industrial control strategies exhibit distinct limitations in the cooperative control tasks of heavy-haul train groups. Although Proportional-Integral-Derivative (PID) control dominated early single-locomotive control due to its structural simplicity[5], its theoretical performance ceiling remains insufficient when satisfying the requirements of multi-agent cooperation within strongly nonlinear environments [6]. PID controllers rely on fixed gain parameters tuned for specific operating conditions, lacking the adaptive capacity to handle system parameter perturbations[7]; consequently, they struggle to cope with dynamic uncertainties introduced by variations in load status, marshaling length, and line gradients [6]. More critically, PID is fundamentally a feedback control mechanism based on current error, lacking feedforward compensation capabilities for the system's large time delays. For instance, when dealing with signal propagation delays in air braking systems that can last tens of seconds, PID control logic fails to effectively suppress longitudinal impulses during the braking wave propagation, easily triggering chain collisions and stretching between vehicles, which causes coupler stress to exceed safety thresholds [6]. To address multi-variable coupling and hard constraint issues, Model Predictive Control (MPC) has become a mainstream research focus in academia in recent years. However, in the process of practical engineering translation, MPC faces two major challenges: computational complexity and model mismatch [8]. On one hand, as an ultra-high-dimensional system, the state space complexity of a heavy-haul train grows rapidly with the number of vehicles; a train comprising 200 wagons involves over 400 state variables. Solving such large-scale, non-convex, and nonlinear optimization problems in real-time on computationally constrained onboard embedded platforms is extremely difficult, making it hard to meet millisecond-level real-time control requirements [8]. On the other hand, the control precision of MPC depends on the accuracy of the internal prediction model. To reduce computational load, existing MPC schemes often employ simplified multi-mass models or linearize the system, ignoring non-smooth gap characteristics between couplers and complex fluid dynamic details [8]. This structural deviation between the simplified model and the physical system leads to significant prediction errors during critical condition transitions, and in extreme environments, it may even generate erroneous instructions that compromise operational safety [8].

Given the real-time performance and applicability defects of model-based methods, Multi-Agent Reinforcement Learning (MARL) offers a novel data-driven paradigm for resolving this problem [9]. The spatial distribution and local interaction characteristics of heavy-haul train groups naturally align with the "Centralized Training, Decentralized Execution" (CTDE) architecture of MARL. Unlike MPC, which relies on high-precision models, MARL possesses model-free characteristics, enabling it to directly extract nonlinear features such as coupler gaps and braking hysteresis through interaction with the environment, thereby circumvent-

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ing biases introduced by modeling simplification. Furthermore, the CTDE paradigm shifts computation-intensive policy optimization to the offline phase, while online control requires only lightweight neural network inference. This compresses instruction generation time to the millisecond level, significantly enhancing the system's responsiveness to transient disturbances.

Despite the theoretical advantages of MARL, its practical deployment remains confronted by three primary challenges [10]. First is the issue of safety assurance: classic RL relies on the "soft constraints" of reward functions, which cannot mathematically guarantee hard physical constraints such as coupler forces. In exploration phases or Out-of-Distribution (OOD) scenarios, this can easily lead to prohibited actions, violating the "Fail-Safe" criterion of industrial control [11]. Second is the Sim-to-Real gap: existing training environments are often constructed based on simplified dynamics[12], making it difficult to reproduce complex wheel-rail contact and time-varying resistance found on real lines, resulting in significant performance degradation or instability of policy models in physical environments [10]. Finally, there is the bias in communication environment assumptions: mainstream algorithms often presuppose ideal communication, ignoring inevitable delays and packet losses in complex electromagnetic environments like mountains or tunnels[12, 13]. Research indicates that cooperative decision-making based on outdated information can trigger longitudinal oscillations in train groups, severely undermining cooperative stability [14].

Addressing the aforementioned issues, this paper proposes a cooperative control enhancement framework for heavy-haul train groups that integrates the mechanism of Artificial Potential Fields (APF). The main contributions of this paper are as follows:

1. The cooperative control problem of heavy-haul train groups is modeled as a Multi-Agent Reinforcement Learning task, utilizing the CTDE paradigm to achieve model-free, distributed real-time cooperative control without the need for precise analytical models.

2. Artificial Potential Field theory is introduced to optimize the reward mechanism, reconstructing discrete safety constraints such as collision avoidance and coupler breakage prevention into continuous, differentiable potential energy penalties, thereby providing agents with explicit gradient information for risk avoidance.

3. A high-fidelity simulation platform containing complex longitudinal dynamic characteristics is constructed, verifying the effectiveness of the introduced potential field mechanism in improving algorithm convergence speed and ensuring operational safety.

The remainder of this paper is organized as follows: Section 1 elaborates on the system modeling of heavy-haul train groups; Section 2 introduces the proposed Multi-Agent Reinforcement Learning method enhanced by potential fields; Section 3 presents experimental results and analysis; and Section 4 concludes the paper.

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1 System Modeling

Constructing a high-fidelity simulation interactive environment and a rational group information topology serves as the foundation for researching the cooperative control of heavy-haul train groups. This chapter aims to establish a system model oriented towards data-driven control from two dimensions: physical evolution mechanisms and information interaction architecture. Specifically, by introducing nonlinear coupler characteristics and complex line resistances, a multi-point mass longitudinal dynamics model is constructed to accurately capture the longitudinal impulse and energy transfer processes under long formation conditions. Furthermore, combined with the actual operation conditions of heavy-haul railways, the communication topology of the train group is established, and the local observation space and action space of the agents are defined, thereby completing the mapping from physical entities to the multi-agent reinforcement learning environment.

1.1 Longitudinal Dynamics Model of a Single Heavy-Haul Train

Distinct from simplified single-point mass models often employed in traditional control research, this section establishes a longitudinal dynamics model based on a multi-point mass system. This approach allows for the accurate capture of critical heavy-haul characteristics, including long formations, large inertia, and complex transmission of coupler forces [15]. Specifically, a heavy-haul train comprising M vehicles (locomotives and wagons) is discretized into a multi-body system connected by nonlinear spring-damper units representing coupler connections[16]. This model accurately simulates internal longitudinal impulses and energy transfer processes, thereby providing a high-fidelity interactive environment for subsequent data-driven algorithms.

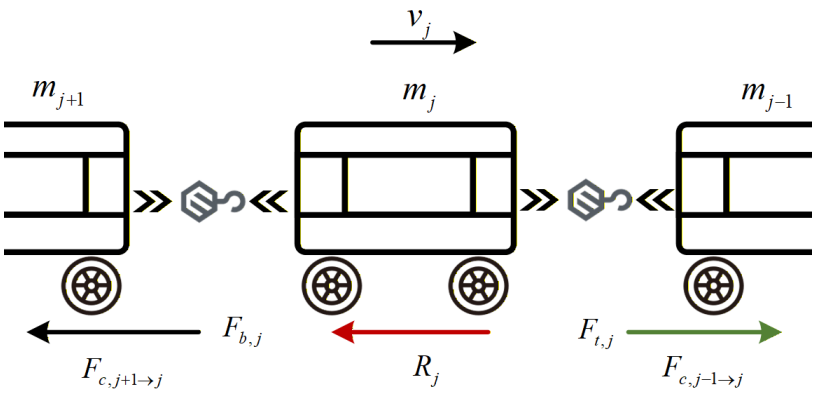


Figure 1: Schematic diagram of force analysis for a single heavy-haul train.

For the j-th vehicle in the train ($j = 1, 2, ..., M$), the longitudinal dynamics equation is

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governed by Newton's second law as follows:

$$m_j \dot{v}_j = F_{t,j} - F_{b,j} - R_j(v_j, s_j) + F_{c,j-1\to j} - F_{c,j\to j+1} \tag{1}$$

where m_j and v_j denote the mass and velocity of the j-th vehicle, respectively; $F_{t,j}$ represents the traction force applied to the locomotive nodes; $F_{b,j}$ indicates the combined braking force considering system delays; $R_j(v_j, s_j)$ is the total running resistance composed of the basic Davis resistance and additional line resistances (gradient and curve); and $F_{c,j-1\to j}$ and $F_{c,j\to j+1}$ denote the internal nonlinear coupler forces acting on the vehicle.

To facilitate high-level decision-making for group cooperation while maintaining simulation fidelity, the dynamics of the entire train are encapsulated into an input-output module. We model the train as a system that receives a normalized total traction/braking command $u_i(t)$ from the cooperative controller and outputs the actual running state, including the macroscopic speed $v_i(t)$ (pilot locomotive or average speed), position $s_i(t)$, and estimated maximum coupler force. Internally, the command $u_i(t)$ is distributed to various power units via a vehicle dynamics actuator, which accounts for multi-locomotive synchronization and air braking wave propagation delays. Consequently, for the agent in the cooperative control layer, the dynamic process of a single train is abstracted as a nonlinear mapping relation $\dot{v}_i(t) = f(v_i(t), u_i(t), \Theta_i, \xi(t))$, where Θ_i contains fixed parameters such as train mass and formation length, and $\xi(t)$ represents time-varying environmental disturbances such as line gradient and aerodynamic drag.

1.2 Group Communication Topology and State Space

The heavy-haul train group consists of N homogeneous or heterogeneous heavy-haul train units. To ensure system robustness and minimize communication bandwidth usage, this study adopts the Predecessor-Following combined with Global Broadcast communication topology, which is widely recognized in engineering practice for its maturity and reliability [17]. Within this topology, the i-th train ($i = 2, \ldots, N$) acquires real-time running state information of its immediate predecessor $i-1$ via Vehicle-to-Vehicle (V2V) communication links [18], while simultaneously receiving the global reference command $v_{\text{ref}}(t)$ from the ground dispatch center via the Train-to-Ground wireless network. As the group leader, the first train ($i = 1$), having no physical predecessor, makes decisions primarily based on global reference commands and line speed limit information, serving as the benchmark to guide the subsequent trains.

Under this topology, the local observation space o_i^t of the i-th agent at time t is defined as a high-dimensional state vector:

$$o_i^t = [v_i(t), \quad a_i(t), \quad \Delta v_{i,i-1}(t), \quad \Delta s_{i,i-1}(t), \quad e_{v,i}(t), \quad \phi(s_i(t))] \tag{2}$$

where $v_i(t)$ and $a_i(t)$ denote the real-time velocity and acceleration of the ego train, respectively; $\Delta v_{i,i-1}(t) = v_{i-1}(t) - v_i(t)$ represents the relative velocity with respect to the preceding train;

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$\Delta s_{i,i-1}(t) = s_{i-1}(t) - s_i(t) - L_{i-1}$ denotes the net spacing, with L_{i-1} being the formation length of the preceding train; $e_{v,i}(t) = v_{\text{ref}}(t) - v_i(t)$ indicates the tracking error relative to the global reference speed; and $\phi(s_i(t))$ is the vectorized encoding of line environment features, including gradient, curve radius, and speed limits at position $s_i(t)$.

The action u_i^t is defined as a continuous scalar representing the normalized traction or braking control command within the range $[-1, 1]$. Specifically, positive values ($u_i^t \in (0, 1]$) map to the traction throttle opening, corresponding to the application of locomotive traction force; conversely, negative values ($u_i^t \in [-1, 0]$) map to the braking level of air or dynamic braking systems. This continuous design enables the agent to generate fine-grained control outputs, thereby effectively mitigating longitudinal impulses associated with discrete notch switching.

2 Data-Driven Multi-Agent Reinforcement Learning Framework

Addressing the inherent strong nonlinear coupling and multi-objective conflicts in the cooperative control of heavy-haul train groups, traditional control methods based on analytical models often fail to balance precision [19], real-time performance, and generalization capabilities. Therefore, this chapter constructs a fully data-driven MARL control framework, transforming the cooperative cruising problem into a sequential decision-making process within a complex dynamic environment. This framework is centered on the MAPPO algorithm and adopts a CTDE architecture to adapt to actual train communication constraints. By designing a composite reward mechanism that integrates potential field safety guidance and energy efficiency optimization, it solves the exploration dilemma under sparse rewards. Furthermore, relying on a high-fidelity simulation environment for closed-loop iteration, the framework achieves autonomous evolution and robustness improvement of cooperative strategies.

2.1 Fundamentals of the MAPPO Algorithm

Given the characteristics of continuous action spaces, high state dimensions, and dynamic environmental changes in heavy-haul train group cooperative control tasks, this paper selects the MAPPO algorithm [20, 21], which has demonstrated superior performance and robustness in multi-agent cooperative tasks. The framework adopts a CTDE architecture, aiming to solve the non-stationarity problem caused by the simultaneous evolution of agent strategies in multi-agent environments, while simultaneously satisfying the requirements of actual train operations for distributed real-time decision-making.

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2.1.1 Algorithm Principles

Under this architecture, each heavy-haul train is treated as an independent agent, and the system comprises two core neural network components working in tandem. The Distributed Execution Policy Network, denoted as π_{θ_i}, operates during the inference or actual operation phase. It receives only the agent's local observation vector o_i^t and maps this input to the probability distribution parameters of the action via a Multi-Layer Perceptron. Subsequently, it samples to obtain the continuous traction or braking control command u_i^t. This design ensures that the train can independently complete cooperative driving tasks relying solely on Vehicle-to-Vehicle communication and local perception.

Conversely, the Centralized Training Value Network, denoted as V_ϕ, is introduced during the training phase to assist the Actor in learning the global optimal policy. This network has access to global state information s^t, including observation information of all trains, global environmental parameters, and unobserved hidden states. The Critic network calculates the Generalized Advantage Estimation by evaluating the value of the current joint state, thereby providing low-variance guidance signals for the gradient updates of each Actor network. Notably, after training is completed, the Critic network is detached and does not participate in online control.

2.1.2 Algorithm Optimization Objectives

The core of the MAPPO algorithm lies in guaranteeing the monotonic improvement of training by limiting the step size of policy updates. For the policy network, its optimization objective is to maximize an objective function with a Clipped term to avoid performance collapse caused by excessive policy update magnitudes:

$$L^{CLIP}(\theta) = \hat{\mathbb{E}}_t \left[\min \left(\rho_t(\theta)\hat{A}_t, \text{clip}(\rho_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t \right) + \sigma S[\pi_{\theta_i}](o_i^t) \right] \tag{3}$$

where $\rho_t(\theta)$ represents the probability ratio of the new strategy to the old strategy, and ϵ serves as the clipping hyperparameter used to constrain this probability ratio within the interval $[1-\epsilon, 1+\epsilon]$. The term $\hat{A}_t$ denotes the GAE function, measuring the advantage degree of the current action relative to the average level. Additionally, $S[\pi_{\theta_i}]$ is the policy entropy term with the coefficient σ, which regulates the balance between exploration and exploitation to prevent premature convergence to local optima. The value network V_ϕ is updated by minimizing the Mean Squared Error between the predicted value and the real return:

$$L^{VF}(\phi) = \hat{\mathbb{E}}_t \left[\left(V_\phi(s^t) - \hat{R}_t \right)^2 \right] \tag{4}$$

where $\hat{R}_t$ is the discounted actual cumulative return. Through the alternating iterative updates of the Actor and Critic, the heavy-haul train group can gradually learn efficient coop-

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erative strategies that satisfy safety constraints and optimize energy efficiency under complex line conditions.

2.2 Reward Function Design

In reinforcement learning frameworks that do not rely on precise analytical models, the design of the reward function dictates the direction of agent policy iteration and the performance boundaries of final convergence. Addressing the multi-objective optimization challenges faced in the cooperative control of heavy-haul train, specifically the need to simultaneously achieve high-precision speed tracking, compact formation maintenance, and minimal energy consumption under the premise of strictly guaranteeing operational safety, a single reward signal is insufficient. Therefore, this paper designs a multi-dimensional composite reward function mechanism, transforming sparse terminal task objectives into dense single-step guidance signals. The total reward r_i^t for agent i at time t is defined as the weighted sum of three sub-reward terms: cooperative tracking, safety potential field guidance, and operational energy efficiency.

2.2.1 Cooperative Tracking Reward

The cooperative tracking reward aims to guide each heavy-haul train agent to strictly follow global dispatch commands at the macroscopic level. This reward term focuses not only on the punctuality of individual trains but also emphasizes the relative state stability among the train group, thereby suppressing the longitudinal multi-point mass oscillation effects caused by the long formation of heavy-haul trains [19]. The specific formal definition is as follows:

$$r_{i,\text{track}}^t = -\left[\alpha_1(v_i^t - v_{\text{ref}}^t)^2 + \alpha_2(d_i^t - d_{\text{desired}})^2 + \alpha_3(v_i^t - v_{i-1}^t)^2\right] \tag{5}$$

where the first term $(v_i^t - v_{\text{ref}}^t)^2$ represents the speed tracking error penalty, guiding the train to closely follow the global reference speed v_{ref}^t to ensure adherence to the schedule and compliance with line speed limits. The second term $(d_i^t - d_{\text{desired}})^2$ denotes the inter-train distance maintenance error penalty, where d_i^t is the actual spacing and d_{desired} is the preset ideal economic cruising distance. This distance is typically set as a fixed or dynamic value slightly larger than the minimum safe braking distance to balance line capacity and operational risks. The third term $(v_i^t - v_{i-1}^t)^2$ is the relative velocity error penalty, introducing a damping mechanism that encourages the ego train to maintain a relatively static state with respect to the preceding train. This is critical for suppressing coupler force impacts caused by frequent acceleration and deceleration. Finally, $\alpha_1, \alpha_2, \alpha_3$ are positive weight coefficients used to adjust the priority among speed tracking precision, spacing maintenance capability, and formation stability.

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2.2.2 Safety Guidance Reward

During the initial phase of reinforcement learning training, the random exploration of agents is highly prone to violating hard safety constraints. Relying solely on simple episode termination or fixed penalties leads to sparse rewards, making it difficult for agents to derive effective collision avoidance strategies. To address this, inspired by Artificial Potential Field methods [22, 23], this paper constructs a repulsive potential function based on relative distance. This transforms discrete safety constraints into continuously differentiable penalty signals. The safety potential function $U(d_i^t)$ regarding the relative net spacing d_i^t is defined as:

$$U(d_i^t) = \begin{cases} \frac{1}{2}\eta\left(\frac{1}{d_i^t - d_{\text{limit}}} - \frac{1}{d_{\text{barrier}} - d_{\text{limit}}}\right)^2, & d_{\text{limit}} < d_i^t < d_{\text{barrier}} \\ 0, & d_i^t \geq d_{\text{barrier}} \\ M_{\text{penalty}}, & d_i^t \leq d_{\text{limit}} \end{cases} \tag{6}$$

Algorithm 1 MAPPO-based Cooperative Control Training Process

Initialize Actor parameters θ_i, Critic parameters ϕ, and Replay Buffer $\mathcal{D}$.
for episode $= 1$ to M_{max} **do**
  Reset environment and obtain initial observation o^0.
  for step $t = 1$ to $T_{episode}$ **do**
    for agent $i = 1$ to N **do**
      Generate action distribution $\pi_{\theta_i}(\cdot|o_i^t)$ and sample action u_i^t.
    end for
    Execute joint action $\mathbf{u}^t = [u_1^t, \ldots, u_N^t]$ in simulation environment.
    Observe next state s^{t+1}, local observations $\mathbf{o}^{t+1}$, and reward r^t.
    Store transition $(\mathbf{o}^t, \mathbf{u}^t, r^t, \mathbf{o}^{t+1})$ in Buffer $\mathcal{D}$.
  end for
  if Buffer size $> B_{min}$ **then**
    for epoch $= 1$ to K_{update} **do**
      Sample mini-batch of trajectories from $\mathcal{D}$.
      Calculate Generalized Advantage Estimation (GAE) $\hat{A}$.
      Update Critic ϕ by minimizing $L^{VF}(\phi)$ (Eq. 4).
      Update Actor θ by maximizing $L^{CLIP}(\theta)$ (Eq. 3).
    end for
    Clear Buffer $\mathcal{D}$ (on-policy).
  end if
  Decay exploration noise variance.
end for

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where d_{limit} represents the physically impassable absolute danger boundary, determined by the relative braking distance model to ensure collision avoidance even under extreme braking conditions. As spacing approaches this boundary, the potential value tends towards infinity (truncated to M_{penalty}), simulating a rigid repulsive force. d_{barrier} serves as the trigger threshold for the safety potential field, creating a buffer warning zone. When $d_i^t \geq d_{\text{barrier}}$, the train is in the absolute safety zone with zero potential, allowing the agent to focus on other objectives. The coefficient η determines the steepness of the potential field gradient. This design provides a smooth yet drastically increasing gradient signal within the buffer zone, enabling agents to perceive impending risks and apply braking in advance, significantly improving exploration efficiency. The final safety guidance reward is:

$$r_{i,\text{safety}}^t = -U(d_i^t) \tag{7}$$

2.2.3 Energy Efficiency and Running Smoothness Reward

Given the "heavy loads and long formations" characteristic of heavy-haul railways, it is essential to incorporate running economy and smoothness into optimization objectives [24, 25]. The reward function $r_{i,\text{econ}}^t$ is designed to suppress oscillations in control inputs and penalize high energy consumption:

$$r_{i,\text{econ}}^t = -\left[\alpha_4 \mathcal{E}(u_i^t, v_i^t) + \alpha_5 \left(\frac{a_i^t - a_i^{t-1}}{\Delta t}\right)^2\right] \tag{8}$$

where the Energy Efficiency Optimization Term $\mathcal{E}(u_i^t, v_i^t)$, typically defined as the square of the normalized control force or instantaneous power, penalizes the magnitude of control actions. This effectively suppresses "sawtooth" driving behaviors and encourages coasting. The Smoothness and Jerk Constraint Term characterizes the rate of change of acceleration. Since jerk is directly correlated with the rate of change of coupler forces in multi-point mass models, explicitly penalizing it forces the agent to learn smooth control curves, ensuring running smoothness and protecting vehicle connecting devices. α_4 and α_5 are the corresponding weight coefficients.

2.3 Training Process

Based on the constructed simulation environment and the MAPPO algorithm, the training process is conducted within an episodic iterative framework. The detailed execution flow is presented in Algorithm 1.

3 Simulation Experiments and Results Analysis

To validate the effectiveness and robustness of the proposed MAPPO-based cooperative control framework for heavy-haul train groups, this section details the construction of a high-

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fidelity simulation platform. The experiments are designed to simulate a scenario involving three heavy-haul trains under complex operating conditions, including starting, acceleration, cruising, and deceleration. The primary objective is to verify the algorithm's capability to achieve precise speed tracking and safe distance maintenance while strictly adhering to physical constraints such as coupler forces.

3.1 Experimental Design

3.1.1 Simulation Setup and Control Objectives

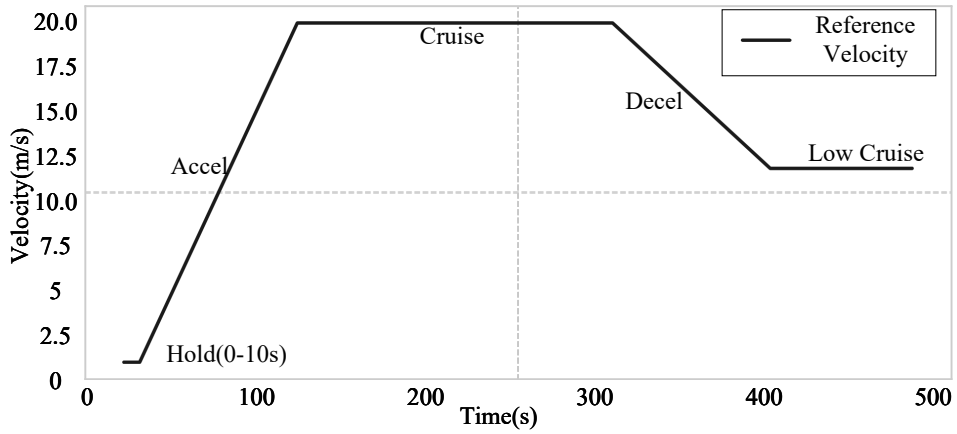


Figure 2: Reference Velocity Profile Design.

The experimental scenario simulates a cooperative formation consisting of $N = 3$ homogeneous heavy-haul trains, denoted as Train 1, Train 2, and Train 3. Unlike simplified single-mass models, this study adopts the multi-point mass longitudinal dynamics model established in Section 1 to accurately capture internal force transmission. Specifically, each train unit consists of one locomotive and 19 heavy-duty wagons. The mass of the locomotive is $150,000$ kg, and each wagon weighs $80,000$ kg, resulting in a total single train mass of $1,670$ t. Regarding dynamics, the connections between vehicles are modeled using nonlinear spring-damper elements to simulate hysteresis effects and slack characteristics.

To ensure formation stability and decouple error propagation, a Virtual Leader strategy within a Predecessor-Following topology is employed. Train 1 acts as the leader and tracks a Virtual Reference Point moving strictly according to the global schedule and optimal speed profile $v_{ref}(t)$ to prevent oscillation amplification at the head of the formation. Meanwhile, Train 2 and Train 3 operate in follower mode, tracking the state of their immediate predecessor via V2V communication while simultaneously referencing the global guidance.

The simulation covers a complete operation cycle to test the dynamic adaptability of the MARL agents, encompassing starting, acceleration, cruising, and braking phases. Throughout these phases, the control objectives are set to minimize the speed tracking error and distance error while strictly ensuring that the maximum internal coupler force remains within the safety

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threshold of ±1000 kN and the inter-train distance never breaches the safety wall defined in Equation (9).

3.1.2 Reference Velocity Profile Design

To comprehensively evaluate the dynamic response and cooperative performance of the heavy-haul train group under varying operating conditions, a specific reference velocity profile $v_{ref}(t)$ is designed for the Virtual Leader as shown in Figure 2. The operation cycle is divided into five distinct phases to rigorously test the agents' ability to handle traction, braking, and state transitions. The specific time intervals, velocity targets, and operational objectives for each phase are detailed in Table 1.

Table 1: Description of Simulation Phases.

Phase	**Time (s)**	**Target Velocity**
Hold	$0 - 10$	0 m/s
Acceleration	$10 - 110$	Ramp to 19.5 m/s
High-Speed Cruise	$110 - 310$	Constant 19.5 m/s
Deceleration	$310 - 410$	Ramp to 11.1 m/s
Low-Speed Cruise	> 410	Constant 11.1 m/s

3.2 Results Analysis

Based on the high-fidelity simulation platform and the reference velocity profile defined in Section 4.1, this section presents a comprehensive comparative analysis of the experimental results. To explicitly verify the effectiveness of the safety guidance mechanism proposed in this paper, the APF-MAPPO (MAPPO integrated with Artificial Potential Field) framework is benchmarked against the baseline Standard MAPPO algorithm. The comparative evaluation is conducted across four critical dimensions: training convergence, tracking precision, operational safety.

3.2.1 Training Convergence Analysis

The evolution of the average episodic reward during the training process of the proposed APF-MAPPO framework is illustrated in Figure 3. The light grey curve represents the raw reward fluctuation for each episode, while the solid blue curve depicts the moving average with a window size of $N = 50$, providing a clearer view of the learning trend.

As shown in Figure 3, the training process over 2,000 episodes evolves through three identifiable phases. Initially, from episode 0 to 500, the curve exhibits low rewards hovering between -8000 and -6000 with high variance, attributed to frequent safety violations during random exploration. A rapid learning phase follows between episodes 500 and 1200, where a sharp upward trend emerges as agents learn to evade potential field penalties and optimize control strategies.

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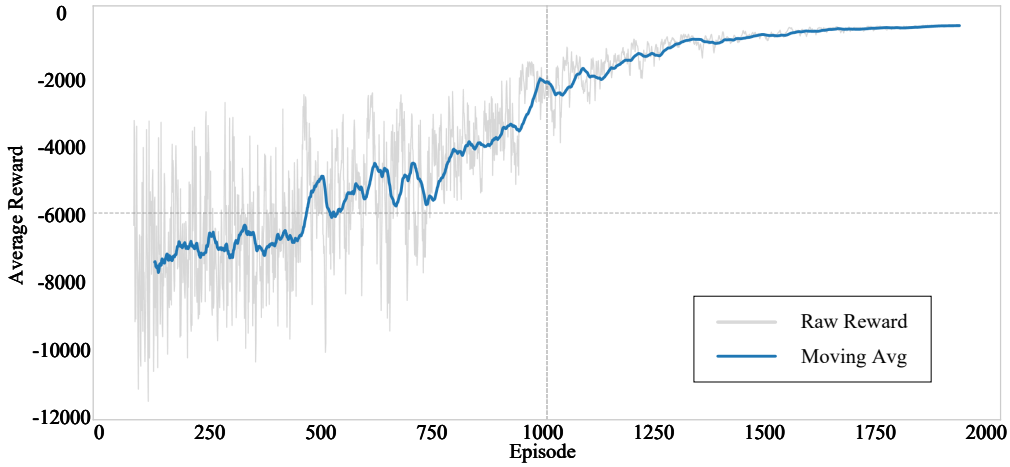


Figure 3: Training Convergence Analysis.

Finally, beyond episode 1200, the reward stabilizes near zero with minimal variance, indicating a successful transition from stochastic exploration to deterministic exploitation. This convergence confirms that the proposed multi-objective reward function provides effective gradient signals, enabling the system to locate the global optimum efficiently.

3.2.2 Tracking Precision and Formation Control Analysis

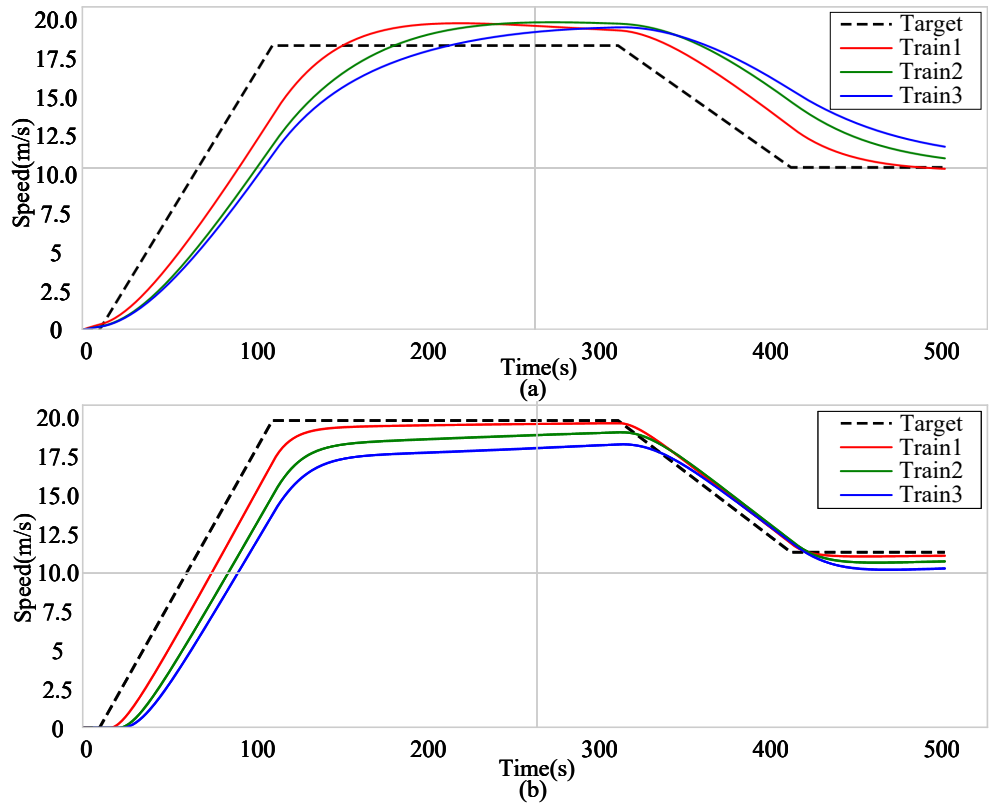


Figure 4: Tracking Precision and Speed Synchronization.

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This section evaluates the cooperative performance from three perspectives: velocity tracking, spatial-temporal consistency, and distance maintenance.

First, to evaluate the dynamic response capabilities, Figure 4 presents the velocity tracking curves. As shown in Figure 4(a), the Standard MAPPO strategy exhibits significant limitations, specifically showing substantial overshoot reaching nearly 21 m/s during the acceleration-to-cruising transition and significant hysteresis during deceleration. The wide spatial distribution between the curves indicates poor synchronization. In contrast, the proposed APF-MAPPO method in Figure 4(b) demonstrates superior fidelity. The velocity curves tightly adhere to the reference trajectory, eliminating overshoot and responding immediately to braking demands. The high clustering of curves proves that the Artificial Potential Field acts as a cohesive force for synchronization.

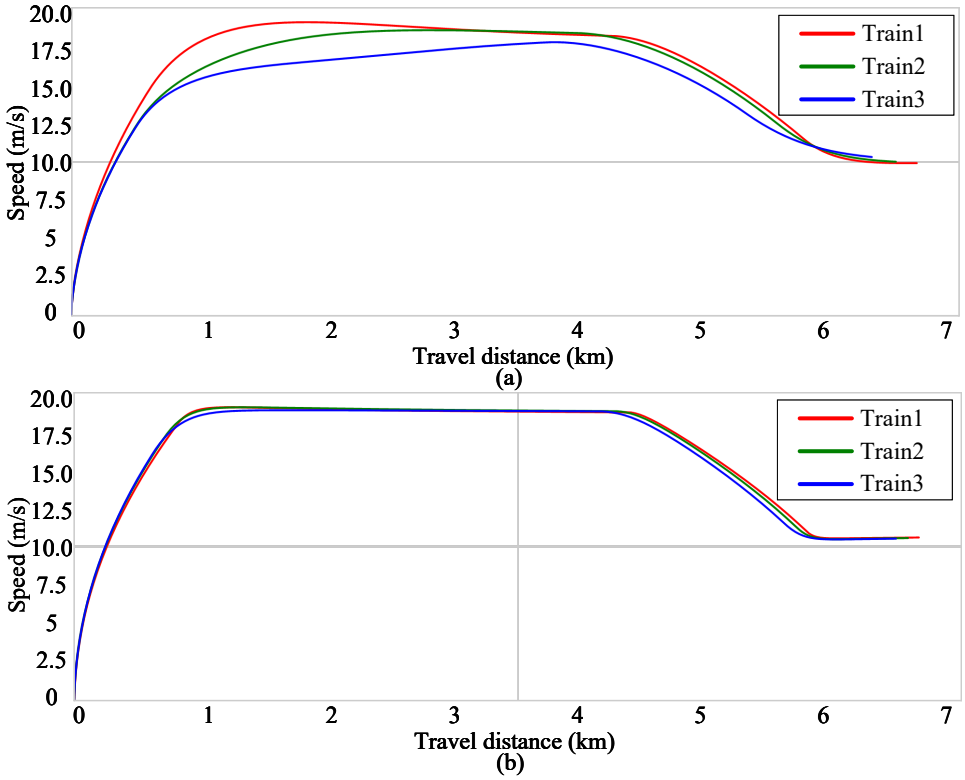


Figure 5: Spatial-Temporal Consistency Analysis.

To further visualize the formation's spatial distribution, Figure 5 plots the Velocity-Position ($v - s$) phase plane. The baseline strategy in Figure 5(a) shows instability, with fluctuating horizontal distances between curves indicative of the "accordion effect." Conversely, the APF-MAPPO method shown in Figure 5(b) demonstrates excellent spatial consistency. The trajectories of followers are almost perfect geometric translations of the leader's trajectory, maintaining a constant horizontal spacing of approximately 2 km and confirming strictly

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maintained cruising speeds.

Regarding formation maintenance, Figure 6 illustrates the time-varying spacing error. The baseline strategy in Figure 6(a) reveals a severe inability to maintain rigidity, with errors accumulating to a peak of 310 m and failing to converge. However, the APF-enhanced strategy shown in Figure 6(b) is significantly more effective. Although an initial deviation occurs during acceleration, the error quickly converges to a stable value of approximately 130 m, representing a 58% reduction compared to the baseline. Crucially, the error curves become flat horizontal lines during cruising, indicating the potential field locks the trains into a fixed relative position.

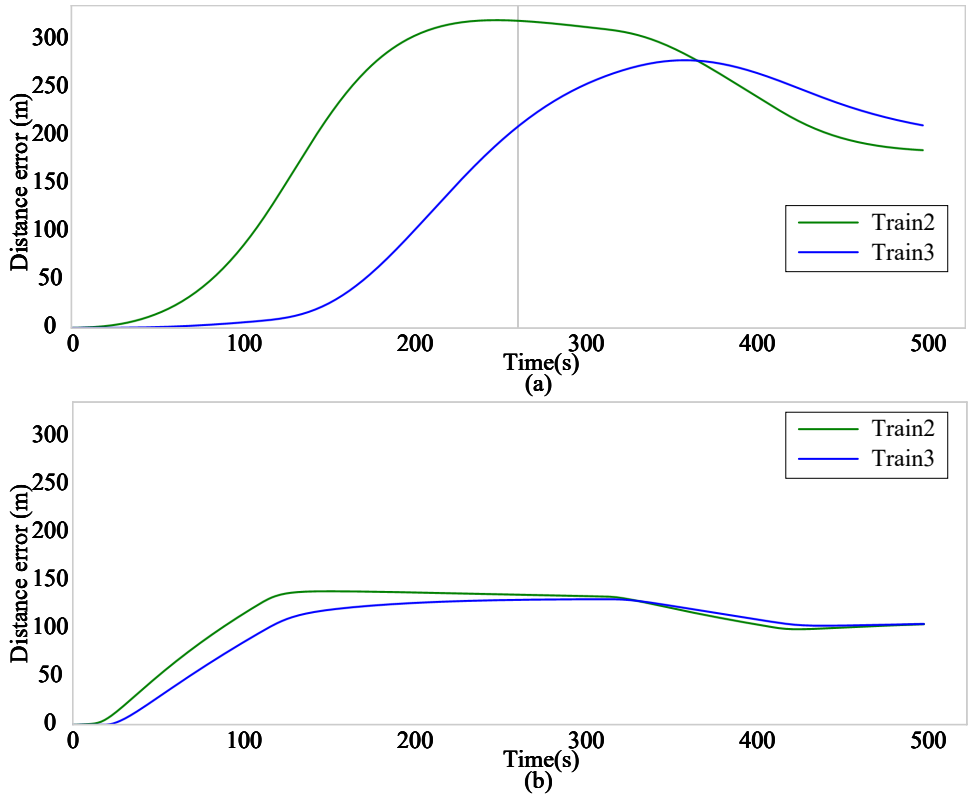


Figure 6: Formation Maintenance and Distance Error.

3.2.3 Operational Safety Analysis

Internal longitudinal coupler force serves as a decisive metric for safety. Figure 7 compares the simulated coupler forces for the two control strategies.

As illustrated in Figure 7(a), the Standard MAPPO strategy exhibits noticeable instability, particularly during the initial starting phase where a sharp oscillation occurs. The force abruptly dips to a compression value of approximately -250 kN before surging to tension, creating a "whiplash" effect that indicates a failure to synchronize acceleration instantly. In comparison, the proposed APF-MAPPO method shown in Figure 7(b) demonstrates superior ride

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comfort. The coupler force profile is remarkably smooth, rising gradually from zero to a peak traction of approximately 300 kN without negative compression spikes during startup. This indicates that the Artificial Potential Field effectively acts as a virtual damper. By eliminating high-frequency oscillations and sudden jerks, the strategy significantly reduces mechanical fatigue and ensures the maximum coupler force remains well within safe limits throughout the operation.

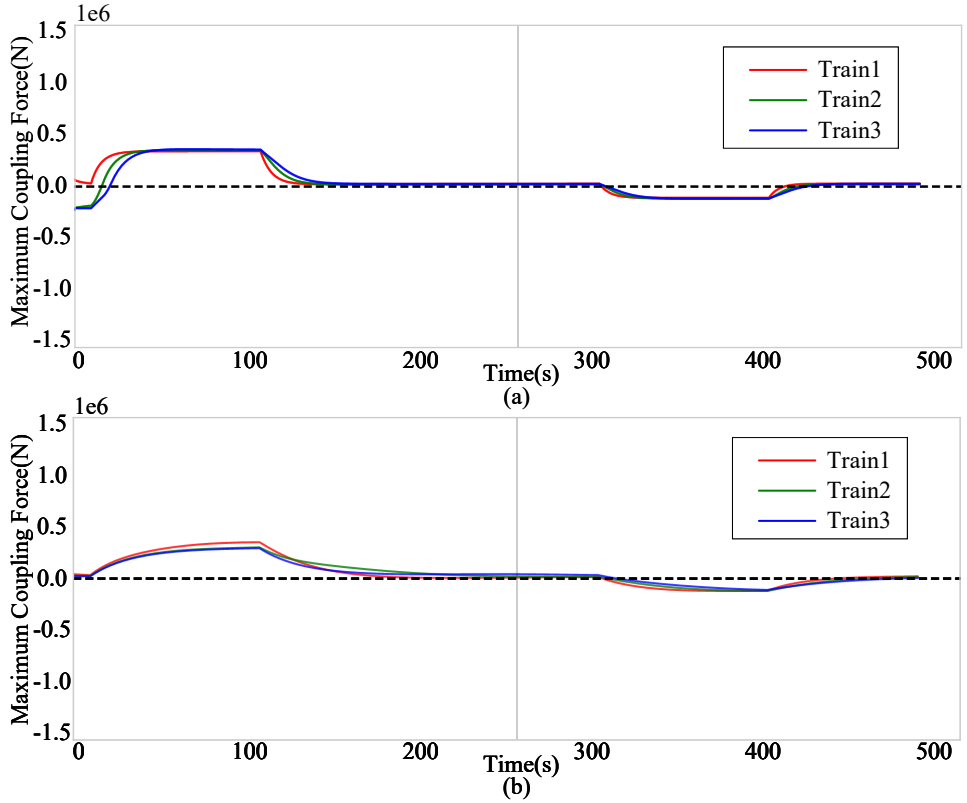


Figure 7: Coupler Force and Ride Comfort.

4 Conclusion

This paper deeply investigated the topic of heavy-haul train group cooperative control, which holds significant engineering importance. Addressing the dependence of traditional model-driven methods on precise mathematical models and their limitations, this paper pioneered and implemented a fully data-driven multi-agent reinforcement learning solution framework. The main work and conclusions include:

1. Established a highly engineered heavy-haul train multi-point mass longitudinal dynamics simulation model, providing a realistic training and testing environment for data-driven algorithms.

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2. Designed a purely data-driven learning framework centered on MAPPO, integrating cooperative tracking, potential field safety guidance, and multi-objective economic optimization.

3. Through systematic simulation experiments and comparative analysis, verified that the comprehensive performance of the proposed framework in ensuring operational safety, improving speed synchronization accuracy, reducing overall energy consumption, and improving running smoothness is significantly superior to traditional methods, and demonstrates excellent adaptive capabilities and robustness.

This study successfully demonstrates the immense potential of data-driven methods in complex industrial cooperative control problems, providing a new technical path for the intelligent upgrade of heavy-haul railway operations.

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